BOLT: From Motion Tracking to Blind Whole-Body Interaction

Anonymous authors

***Abstract*— Human motion specifies behavior but may assume support surfaces unavailable to a humanoid robot. We study blind support-mismatched motion execution: realizing these commands across different support conditions without explicit scene observations. We present BOLT, a behavior foundation model that learns blind whole-body interaction from synchronized human–robot–support data and privileged support supervision. Support-consistent motion pairing (SCMP) synthesizes missing human commands for robot–scene records, preserving timing while accounting for supporting geometry. Privileged support supervision couples motion imitation with placement, loading, and supported progress during training; deployment uses Skinned Multi-Person Linear (SMPL) motion commands and proprioceptive feedback. Across three stair heights and perturbed starts, one controller completes 80/90 simulated stair trials driven by a held-out level-ground human recording, and 29/30 continuous seated-to-standing, walking, and stair-ascent trials. Additional contacts are reported separately. Offline evaluation shows improved support consistency of synthesized commands. Flat-ground tests execute all ten robot references without falls, with a mean root-relative joint-position error of 34.77 mm. Hardware demonstrations cover sitting, rising, and stair ascent; an all-SMPL simulation extension adds stair descent and table climbing. We observe emergent cross-support execution: level-ground human commands drive stair ascent without that command–execution pairing in training. Project page: https://anonymous.4open.science/w/BOLT-2146/index.html.**

***Key words*— Humanoid robots, behavior foundation models, whole-body control, motion imitation.**

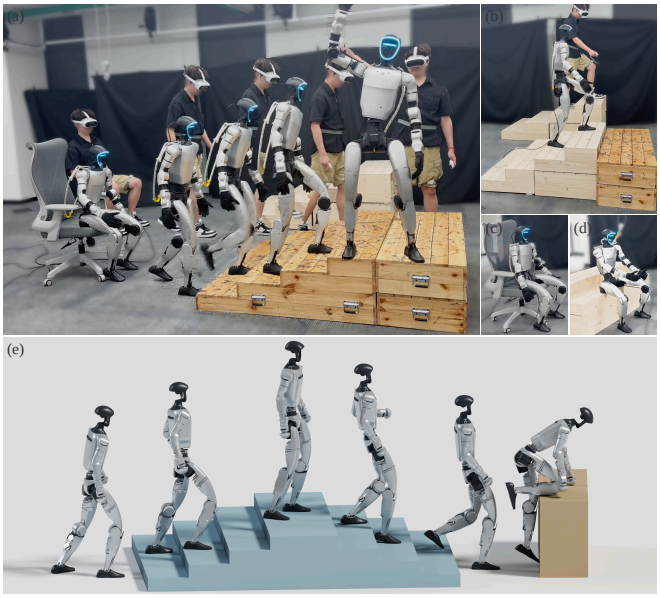


Fig. 1. **Blind whole-body execution across support conditions.** (a) The human and robot rise from sitting; the human walks on level ground while the robot ascends stairs and follows an arm gesture. (b) Separate stair ascent. (c,d) Sitting on an office chair and a rigid support. (e) Simulation task illustration: BOLT executes continuous walking, stair ascent/descent, and table climbing using SMPL commands throughout (Sec. IV-D). Temporal composites illustrate successive support modes.

I. Introduction

Behavior foundation models (BFMs) learn reusable humanoid behaviors [1], with shared human–robot motion interfaces enabling scalable tracking [2]. Motion commands specify activities but underdetermine the support needed in another environment. We study *support-mismatched motion execution*: realizing human motion commands when robot support surfaces differ from those associated with the human motion. We focus on blind execution: the deployed controller receives motion commands and proprioception without explicit scene observations.

Perceptive BFM [3] addresses terrain mismatch through a blind teacher tracking terrain-adapted references and a deployed terrain-conditioned student. PGMT [4] combines terrain glimpses with tracking relaxation. Both condition deployment on online geometry. GigaBrain-WBC-0.5 [5] also demonstrates blind human-command interaction and responses to missing supports, using motion-derived training geometry and a behavior world model. BOLT studies a complementary formulation: synthesizing missing human commands for robot–scene records and learning privileged support objectives for execution across support changes. Table I summarizes reported capabilities.

Consider a seated human and humanoid. Both rise; the human walks on level ground while the robot walks and ascends stairs (Fig. 1). The robot must release seat support and establish footholds the operator never encounters. Climbing onto furniture further requires upper-body support. Establishing, maintaining, and releasing support connects these tasks.

To this end, we present BOLT, a SMPL-conditioned BFM that learns from paired interaction data and privileged support supervision. We adopt the motion-token and control architecture of SONIC [2]. Robot interaction records provide motion and recoverable support geometry but may lack human commands. Our construction aligns available human records and uses support-consistent motion pairing (SCMP) to synthesize missing counterparts. Privileged support supervision complements imitation: region and phase annotations guide where and when to contact, while simulated forces supervise loading and supported progress. Motion commands and proprioceptive feedback drive deployment, without online terrain maps or an explicit geometry predictor.

We evaluate one checkpoint with two held-out human recordings and fixed robot references. Training includes stair

motions but never pairs level-ground human commands with robot stair ascent. We call execution of this untrained pairing *emergent cross-support execution*.

Our contributions are:

- We formulate blind support-mismatched motion execution and introduce BOLT, which realizes human commands across differing robot support geometries without explicit scene observations.
- We construct synchronized human–robot–support data, using SCMP to synthesize missing human-command counterparts and make robot–scene interaction records without human capture usable for motion-conditioned BFM training.
- We introduce privileged support supervision for contact establishment, loading, and release while retaining a blind motion interface. We demonstrate continuous seat-to-foot transitions and stair ascent, and extend this formulation to hand-supported climbing.

II. Related Work

A. Reusable Motion Interfaces

Large motion corpora [6] and physics-based imitation [7], [8] underpin reusable whole-body control. Masked motion priors [1], [9] support partial commands, while whole-body controllers and teleoperation systems [10]–[13] connect human inputs to robot execution. SONIC [2] further aligns human and robot commands through a shared motion representation. BOLT inherits this representation and control architecture, studying how commanded motion is realized when human and robot support conditions differ. MotionWAM [14] generates whole-body motion tokens from visual world-model features. Such high-level generation is complementary to BOLT's focus on physical execution under differing support conditions.

B. Constructing Motion–Scene Correspondence

Interaction data require correspondence between body motion and supporting geometry. Kinematic retargeting [15] maps motion across embodiments; interaction-aware optimization extends this process to preserve body–object–terrain relationships during retargeting and augmentation [16]. Reconstruction and generation offer additional sources: VideoMimic [17] recovers motion and terrain from video, while GRAIL [18] combines known 3D assets with generated video priors to construct interaction references. Physics-based refinement can improve imperfect demonstrations before distilling a shared controller [19]. These routes construct or refine references for execution. SCMP addresses a complementary data requirement: robot–scene records may lack the synchronized human counterpart needed by a shared human–robot interface. It fits human commands while retaining the original robot trajectory, scene, and timestamps, adding the missing command modality to existing interaction records. Support and clearance constraints guide the synthesized human motion toward the source interaction, making these records usable as paired training examples.

C. Contact Commands and Training Supervision

Contact-guided controllers explicitly specify intended interaction. WoCoCo [20] uses prescribed contact sequences, while SceneBot [21] and ContactMimic [22] condition motion tracking on contact labels. These labels let the command distinguish similar poses with different contact intentions; ContactMimic explicitly trains this distinction using augmented trajectories. SceneBot additionally uses global root-state estimates and demonstrates continuous terrain and object interaction. BOLT places support annotations in training supervision: source and phase information select foot, seat, and hand-support objectives, while the deployed actor receives motion commands and proprioception. The distinction concerns where support intent enters the controller, with BOLT learning its realization from paired interaction data and subsequent physical feedback.

D. Perceptive Adaptation and Blind Execution

Perceptive controllers use online geometry to adapt execution. GRAIL's terrain/chair tracker receives terrain observations [18]. PHP [23] combines motion matching with depth-conditioned obstacle traversal. Perceptive BFM [3] constructs paired raw and terrain-conformal references, trains a blind teacher, and distills a student receiving raw commands and local terrain observations. PGMT [4] instead samples motions and terrains independently under compatibility rules, combining terrain glimpses with relaxed lower-body tracking. BOLT's correspondence explicitly aligns human commands, robot references, and support annotations; geometry and support supervise training, while proprioceptive feedback closes the deployment loop without online scene observations.

Blind locomotion and motor adaptation [24], [25] establish that proprioceptive feedback can accommodate changing conditions. GigaBrain-WBC-0.5 [5] is a closely related blind human-command controller. It reconstructs spatial supports from retargeted motion and jointly predicts actions, future proprioceptive states, and latent-command distributions used for command filtering. Both methods use motion-linked training geometry and demonstrate interaction without online scene observations; blind deployment and motion–scene pairing are shared ingredients. BOLT focuses on the missing human-command modality: SCMP synthesizes synchronized commands for robot–scene records while retaining their robot trajectories and timing. Source/phase-specific support objectives train execution, and deployment does not use a predicted command-distribution filter. Our evaluation holds commands fixed across changes in robot support geometry. Table I compares reported capabilities, rather than performance under a shared benchmark.

III. Method

Human and robot support geometries satisfy $\mathcal{G}^{\mathrm{h}} \neq \mathcal{G}^{\mathrm{r}}$; support modes specify supported body parts. BOLT maps motion history $\mathbf{c}_t^m$ and proprioception $\mathbf{o}_t$ to actions (Eq. (6)). Identical observed histories cannot distinguish geometries before interaction; training shapes subsequent feedback.

TABLE I

REPORTED CAPABILITIES OF WHOLE-BODY CONTROLLERS.

Method	Diverse tracking	Human teleop.	Terrain interaction	Object support	Adaptive stairs	Climbing	Scene-free actor
SONIC [2]	✓	✓	×	×	×	×	✓
TWIST [12]	✓	✓	×	×	×	×	✓
OmniRetarget [16]	×	×	✓	✓	×	✓	✓
PHP [23]	×	×	✓	✓	×	✓	×
SceneBot [21]	✓	×	✓	✓	×	×	✓
GRAIL [18]	✓	×	✓	✓	×	×	×
Perceptive BFM [3]	✓	✓	✓	×	✓	×	×
PGMT [4]	✓	✓	✓	×	✓	×	×
GigaBrain-WBC-0.5 [5]	✓	✓	✓	✓	×	×	✓
BOLT	✓	✓	✓	✓	✓	✓	✓

✓ / ×: capability reported / not reported under these definitions; × does not establish inability. Human teleoperation uses causal operator motion without externally supplied contact prompts. Object support includes seated or hand-supported interaction. Adaptive stairs uses commands not matched to robot-side stair geometry. Climbing uses hands or body support to mount raised obstacles. Scene-free excludes explicit terrain/object geometry in the deployed actor. Evaluation settings differ across works.

A. Motion–Scene Data Construction

The pipeline (Fig. 2) combines synchronized human and robot trajectories, supporting geometry, reference contacts, and source category:

$$\mathcal{D} = \left\{ \left(\tau_i^{\mathrm{h}}, \tau_i^{\mathrm{r}}, \mathcal{G}_i, \overline{\mathbf{C}}_i, d_i \right) \right\}_{i=1}^{N}. \tag{1}$$

Here N counts clips; τ_i^{h} and τ_i^{r} are human and robot trajectories, $\mathcal{G}_i$ is the robot scene, $\overline{\mathbf{C}}_i$ denotes intended contacts, and d_i identifies flat, stair, chair, or climbing sources. Physical forces are measured separately in simulation.

Climbing annotations specify a body patch (which), a confidence-weighted phase window (when), and a finite object-relative region (where). They guide contact without fixing one contact point; missing annotations do not forbid contact.

Footfalls and stable support heights determine stair profiles, including skipped levels; stable pelvis support defines an oriented seat cuboid. Shared timestamps, cropping, and coordinates align available human records with robot trajectories. Direction-based inverse kinematics converts other skeletons to compatible human poses.

Toe, heel, and swept-foot clearance checks refine synchronized motions and geometry. Mirroring and reversal preserve contact identities. Validation checks frame counts, joint order, coordinates, support gaps, sliding, and penetration; variants of each source must share one source-disjoint partition.

We draw on BONES-SEED [26], LaFAN1 [27], AMASS [6], and robot–scene records from OmniRetarget [16]. Table II summarizes the available corpus for the climbing extension.

TABLE II

TRAINING CORPUS FOR THE CLIMBING EXTENSION.

Task	Clips	Duration (h)	Sampling
Flat	131,311	272.197	60%
Stair	2,528	7.891	20%
Chair	1,028	2.669	5%
Climb	250	0.816	15%
Total	135,117	283.573	100%

Scope. Counts and durations include augmentation. Sampling denotes configured source weights.

B. Support-Consistent Motion Pairing

Robot–scene records without human capture lack the human branch's required command. SCMP synthesizes that command while preserving its timing and correspondence with robot support. With a fixed human skeleton, we map robot points and supporting boxes into a human-scale scene by $T(\mathbf{p}) = \mathbf{o} + s(\mathbf{p} - \mathbf{o})$, where s is the neutral human-to-robot stature ratio and $\mathbf{o}$ the first robot root's ground projection. The original robot trajectory, scene, and timestamps remain unchanged.

Root orientation follows the robot. Bounded inverse kinematics optimizes human root translation and joint coordinates $\mathbf{x}_t$:

$$\min_{\mathbf{l}_t \le \mathbf{x} \le \mathbf{u}_t} \sum_{k \in \{\mathrm{kin,sup,scene,reg}\}} \|\mathbf{r}_{t,k}(\mathbf{x})\|^2. \tag{2}$$

The weighted kinematic residuals match robot-derived endpoints, root position, segment directions, and orientations. Heel/toe targets retain canonical human foot dimensions. Source foot proximity and vertical speed activate support-height and footprint objectives; consecutive frames on the same surface penalize excess tangential displacement, $[(\mathbf{f}_t - \mathbf{f}_{t-1}) - (\bar{\mathbf{f}}_t - \bar{\mathbf{f}}_{t-1})]_{xy}$, for fitted and target foot midpoints $\mathbf{f}, \bar{\mathbf{f}}$. Source knee/pelvis intervals add positional targets. Scene residuals penalize clearance violations; posture and temporal priors preserve the source-derived initialization and previous fit, advancing only its root by the reference displacement.

Joint, root, and rate bounds constrain four forward fitting passes with bounded Gauss–Newton steps. Between passes,

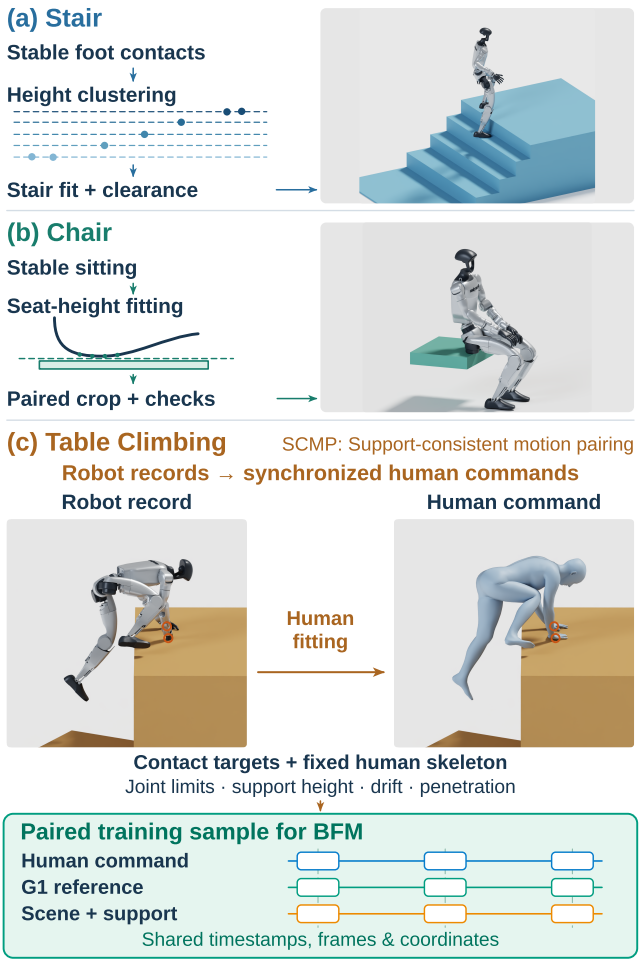


Fig. 2. **Paired motion–scene data construction for BFM training.** (a) Foot-contact heights reconstruct stairs. (b) Stable seated phases determine seat geometry. (c) SCMP fits human commands to robot climbing records with support and scene constraints; orange circles mark palm targets. Outputs align human commands, G1 references, and scene/support annotations. Poses illustrate constructed references.

skinned-mesh penetration checks add body-attached collision probes. Support and clearance are soft objectives; the same mesh inspection contributes to final quality checks (Sec. IV-C).

C. Privileged Support Supervision

Scenes, reference contacts, and simulated forces supervise training while remaining outside actor inputs. Source and phase masks route support objectives alongside inherited tracking and regularization r_t^{base}:

$$r_t = r_t^{\text{base}} + \sum_{d\in\{\text{st,seat,cl}\}} m_t^d r_t^d - w_{\text{u}} c_t^{\text{u}}. \tag{3}$$

The masks select stair sources, chair-source reference-labeled seated phases, and climbing sources, respectively; c_t^{u} counts penalized body contacts and is zero for climbing.

a) Foot and seat support: Gaussian stair rewards track world-frame toe/heel positions and velocities, averaging normalized squared errors across points before exponentiation. Seat rewards track pelvis position and velocity, with separate horizontal/vertical tolerances. An additional indicator detects upward pelvis–seat normal force above 5 N over a three-sample history; it detects contact rather than load sharing.

Reference pelvis-contact labels above 0.5 activate seat rewards and exempt the pelvis from the generic contact penalty; outside seated phases, its usual penalty applies. Ankles, wrists, and elbows are exempt, with $w_{\text{u}} = 0.1$ and a 1 N threshold. Climbing disables this generic penalty throughout the episode to permit upper-body support; physical collisions remain active.

b) Region-guided climbing support: Climbing combines region approach, sustained hand loading, supported progress, and slip control. For hand i, $c_i \in [0,1]$ combines confidence and phase membership. Reach reward R_t^{reach} maximizes valid-hand proximity, decreasing with normal separation and distance outside the region. Proximity receives no load credit.

Let $C(x) = \text{clip}(x,0,1)$, $[x]_+ = \max(x,0)$, and $W = mg$. A hand qualifies for support ($h_i = 1$) after 0.10 s of continuous target-face contact within the region tolerance, sufficient normal compression, and upward force. With relative tangential speed v_i, speed scale v_0, and target-object upward force $F_i^{\uparrow}$, we define

$$\begin{aligned} a_i &= h_i c_i C\left(\frac{[F_i^{\uparrow}]_+}{0.10W[1+(v_i/v_0)^2]}\right), \\ S_t &= \min\left(\sum_i a_i, \max_i h_i c_i\right). \end{aligned} \tag{4}$$

The cap preserves annotation confidence. Forces are control-period means filtered by target object, excluding unrelated contacts.

Within each annotated stage, normalized root progress $s_t \in [0,1]$ projects onto a fixed object-relative start–goal direction. Its running maximum b_t updates even without support:

$$\begin{aligned} b_t &= \max(b_{t-1}, s_t), \\ R_t^{\text{prog}} &= \min(S_{t-1}, S_t) C\left(\frac{b_t - b_{t-1}}{v_{\text{p}}\Delta t}\right). \end{aligned} \tag{5}$$

Here v_{p} scales progress per step Δt. Resets/stage changes initialize $b_t = s_t$ without credit; static stages disable progress. Revisiting attained progress earns nothing. Slip cost C_t^{slip} averages bounded squared excess tangential speed over both hands during target contact, including outside the region; relocation windows can relax it. We use $r_t^{\text{cl}} = 2R_t^{\text{reach}} + 6S_t + 4R_t^{\text{prog}} - C_t^{\text{slip}}$.

D. Motion Interface and Blind Deployment

BOLT accepts SMPL-derived human features [28] or G1 references through a shared motion interface (Fig. 3). For modality $m \in \{\text{h},\text{r}\}$, encoder E_m feeds a shared finite scalar quantizer Q (FSQ) [29] and controller π_θ:

$$\mathbf{z}_t = Q(E_m(\mathbf{c}_t^m)), \qquad \mathbf{a}_t = \pi_\theta(\mathbf{o}_t, \mathbf{z}_t). \tag{6}$$

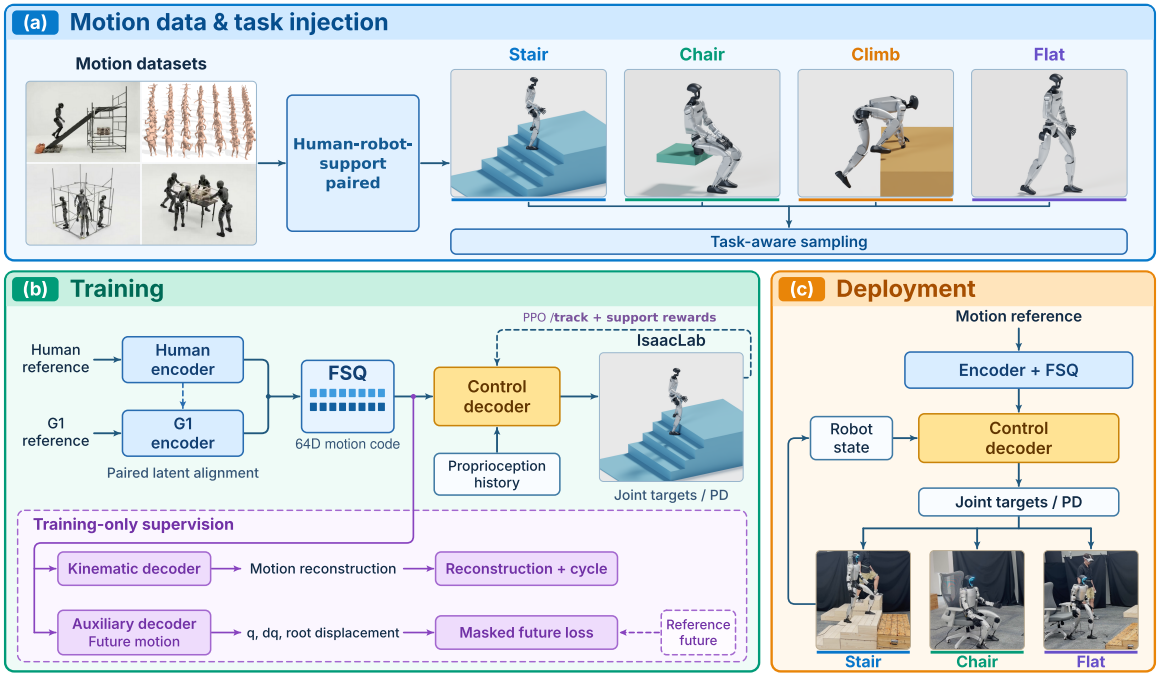


Fig. 3. **BOLT's data, training, and deployment pipeline.** (a) Paired motion–support data with task-aware sampling. (b) Human or G1 commands share a 64D FSQ representation and controller. Representation losses supervise motion encoding, while tracking and privileged support rewards train execution. Future-motion prediction provides auxiliary supervision for the climbing extension. (c) Deployment uses motion commands and proprioceptive feedback, without scene observations or auxiliary decoders.

The proprioceptive history contains joint positions/velocities, angular velocity, gravity direction, and previous actions.

Ten causal human frames at 50 Hz encode local joint positions, relative root orientation, and kinematically mapped wrist angles. The 840D command yields a 64D quantized code; the controller combines it with 930 proprioceptive features to output 29 joint actions. MuJoCo deployment runs inference at 50 Hz with scaled, clipped offsets from the nominal pose, joint/torque limits, and low-level PD control. No robot-reference action assistance or scene geometry enters human-command evaluation.

G1 reconstruction supervises motion content; latent alignment and a re-encoding cycle loss align human/robot commands. The climbing extension additionally predicts motion over 0–2 s. These objectives supervise the shared representation during training; auxiliary decoders are removed at deployment.

IV. Experiments

We test blind execution across support geometries (RQ1), SCMP command quality (RQ2), and continuous support transitions (RQ3). Table III summarizes execution; flat-ground tracking and fixed-target replay examine retention and feedback.

A. Setup and Evaluation Scope

For controller comparisons, we train BOLT from scratch in one stage on flat, stair, and chair motions using proximal policy optimization (PPO), 48 NVIDIA A100 GPUs, and 393,216 environments. The 52,000 iterations take 139.09 h (6,676.11 GPU-hours), from startup to saving. Randomization covers friction, mass/center of mass, default joint angles, support placement, velocity perturbations, and observation noise.

We compare released SONIC and GRAIL, using one checkpoint per method. Quantitative human-command and G1 stair tests share MuJoCo and the G1 asset with capsule hand collisions. Flat tracking uses native IsaacLab (50 Hz control, 200 Hz physics). GRAIL receives actual 121D terrain and 99D object features on stairs. Hardware inference uses C++/TensorRT 10.13.3.9 at 50 Hz on a Ryzen 7 8845H and RTX 4060 Laptop (8 GB).

Quantitative human-command evaluation uses two recordings excluded from training: the continuous seated-to-standing and level-ground walking recording, and a human stair recording captured with PICO and converted to SMPL. Each condition has ten paired starts: one nominal and nine fixed perturbations within ± 2 cm position, $\pm 2^\circ$ yaw, and ± 0.03 m/s horizontal velocity. Human tests perturb world coordinates; G1 stair tests perturb stair-aligned coordinates. Reference and initial-state protocols are fixed within each group.

Completion requires the endpoint and stable target support throughout the final second, with tilt $< 35^\circ$ and horizontal speed < 0.15 m/s. For stairs, both feet must support the fourth tread or its same-height extension at ≥ 20 N each, with root progress to that tread and $|y| < 0.63$ m. Sitting requires pelvis–seat support ≥ 5 N; rising releases it below 5 N and establishes foot support with root height > 0.65 m. Falls require tilt $> 60^\circ$ for 0.1 s; $|y| > 0.80$ m and numerical/state-

TABLE III

EXECUTION ACROSS MOTION INPUTS AND SUPPORT CONDITIONS.

(a) Stair execution: mean progress and outcome rates (%)

		5 cm				10 cm				15 cm				Overall	
Motion command	Method	P	SR	SR_0	FR	P	SR	SR_0	FR	P	SR	SR_0	FR	P	SR
Human flat walking	SONIC	73.3	0	0	83.3	16.7	0	0	100	1.7	0	0	100	30.6	0
	BOLT	**99.2**	**90**	**86.7**	**3.3**	**100**	**96.7**	**83.3**	**3.3**	**90.8**	**80**	**3.3**	**10**	**96.7**	**88.9**
Human stairs ($\Delta x = 0$ cm)	SONIC	75	0	0	0	2.5	0	0	100	0	0	0	80	25.8	0
	BOLT	**95**	**50**	**50**	0	**75**	0	0	**0**	**75**	0	0	**0**	**81.7**	**16.7**
Human stairs ($\Delta x = 10$ cm)	SONIC	75	0	0	0	0	0	0	100	5	0	0	80	26.7	0
	BOLT	**100**	**100**	**100**	0	**92.5**	**30**	**30**	**0**	**72.5**	0	0	**20**	**88.3**	**43.3**
Human stairs ($\Delta x = 20$ cm)	SONIC	75	0	0	100	5	0	0	100	2.5	0	0	100	27.5	0
	BOLT	**100**	**100**	**100**	**0**	**100**	**100**	**100**	**0**	**82.5**	0	0	**10**	**94.2**	**66.7**
G1 stair reference	SONIC	62.5	0	0	0	50	0	0	0	32.5	0	0	60	48.3	0
	GRAIL	75	0	0	0	62.5	0	0	0	25	0	0	10	54.2	0
	BOLT	**100**	**100**	**100**	0	**100**	**100**	**100**	0	**100**	**100**	0	**0**	**100**	**100**

(b) G1 joint tracking: RMSE (rad)

	Step height		
Method	5 cm	10 cm	15 cm
SONIC	0.1679 ± 0.0048	0.1822 ± 0.0093	0.5415 ± 0.0523
GRAIL	0.1301 ± 0.0030	0.1350 ± 0.0057	0.1742 ± 0.0923
BOLT	$\mathbf{0.1139} \pm 0.0050$	$\mathbf{0.0915} \pm 0.0012$	$\mathbf{0.0900} \pm 0.0003$

(c) Seat and continuous transitions (%)

	SONIC		BOLT	
Task	SR ↑	SR_0 ↑	SR ↑	SR_0 ↑
Sitting	33.3	0	**100**	0
Independent rising	100	0	100	**86.7**
Sequence (31.3 s)	0	0	**96.7**	**30**

P: mean highest supported tread / four treads (↑); tread 3 gives 75%. We use foot support ≥ 20 N up to first detected fall, with no dwell requirement. SR: success rate (↑); SR_0: success without extra contact (↑); FR: fall rate (↓). All trials count: $n = 30$ per flat-command height and seat/sequence task, $n = 10$ otherwise. Overall pools the three heights. Both human recordings are excluded from training; Human stairs uses original-speed PICO-to-SMPL capture, with forward start shift Δx. Each input is fixed across methods and heights. RMSE covers all 29 joints and full episodes (mean $\pm$ sample SD). Seat support is permitted; rising starts independently; the sequence has no reset.

clock errors also invalidate completion. Contact-qualified completion additionally requires total non-target environment normal contact force < 5 N throughout; foot and intended pelvis–seat support are allowed. All initiated trials remain in the denominator.

B. RQ1: Blind Execution across Support Geometries

a) Emergent ascent from flat walking: Training never pairs level-ground human commands with robot stair ascent. We replay the fixed command over four-step stairs with 5/10/15 cm risers, 30 cm treads, and 0.90/1.00/1.10 m fronts. Each 14.8 s rollout includes a shared 3 s preparation prefix. Across 30 trials per height (Table III(a)), BOLT achieves 96.7% mean stair progress and 88.9% SR, versus 30.6% and 0% for SONIC. At 15 cm, additional contact lasts a median 14 ms among completed trials (maximum 86 ms).

b) Held-out human stair command: The human stair command is replayed at original speed, with shared endpoint holds giving 7.08 s episodes. The nominal root-to-first-riser distance is 37.21 cm. After the original alignment test, we additionally shift all starts forward by 10 and 20 cm; Table III(a) reports the entire sweep. At $+20$ cm, BOLT completes all 5/10 cm trials without non-foot contact. No human-stair trial completes 15 cm stairs, and completion depends on starting alignment. SONIC completes none across the nine conditions.

c) Fixed G1 stair command: We also replay one G1 stair reference from BOLT's training set unchanged across heights, starting from its frame-3 state for 6.82 s. BOLT completes all 30 trials with no falls, versus no completions for SONIC or GRAIL (Table III(a)). It also has the lowest full-episode 29-joint RMSE (Table III(b)). At 15 cm, every BOLT trial has a 4–10 ms right-shin contact with the second step, so contact-qualified completion is 0/10. These results describe the stated MuJoCo protocols.

C. RQ2: Quality of Synthesized Human Commands

We compare SCMP with kinematic fitting using one original-height record from each of 28 climbing source families in the fixed corpus (17,626 frames). Both share inputs, initialization, kinematic/prior weights, bounds, and optimization budgets. The baseline omits support/scene objectives, scene-conditioned and knee/pelvis targets, and mesh repair. Source human recordings are used only for scoring.

Robot-derived support masks are fixed across methods. We measure absolute skinned-foot support gaps, fixed-foot-midpoint horizontal speed during same-surface stance with source speed ≤ 0.10 m/s, and per-frame maximum mesh penetration. Human fit measures 14 hip-centered joints after uniform limb-length normalization. Gaps and penetration decrease for all 28 sources (Table IV): 22/28 clips pass quality checks versus 0/28; both export all pairs. Sliding and human fit error increase, showing a construction–fidelity tradeoff independently of controller success.

D. RQ3: Continuous Transitions between Support Modes

a) Establishing and releasing seat support: Table III(c) uses seat heights of 29.05/34.05/39.05 cm, ten paired starts

TABLE IV

OFFLINE SCMP CONSTRUCTION QUALITY.

Metric	Kinematic	SCMP
Support gap (mm) ↓	19.61 [17.36, 21.82]	**4.95** [4.66, 5.24]
Foot sliding (cm/s) ↓	**6.38** [5.79, 7.06]	7.34 [6.39, 8.50]
Penetration (mm) ↓	63.95 [48.79, 78.26]	**4.32** [2.92, 5.98]
Human fit error (mm) ↓	**75.99** [69.77, 83.16]	77.47 [71.40, 84.45]
Quality-passing clips ↑	0/28 [0.0, 12.1]	**22/28** [60.5, 89.8]

Values average 28 clips equally: per-clip p95 for the first three metrics and mean for human fit. Brackets give 95% intervals from 10,000 paired source-bootstrap resamples; the last row uses Wilson intervals in percent. Quality requires structural/anatomical checks and all-frame limits: source-supported gap/overlap error and mesh penetration ≤ 20 mm; sole/palm/active-knee target error $\leq 100/70/100$ mm; root offset ≤ 350 mm; local joint speed ≤ 25 rad/s. Sliding and human fit are separate metrics.

each, and shared 2 s final-pose holds. Rising starts from a valid seated state independently of sitting. BOLT completes both phases throughout; SONIC completes sitting only at the highest seat, but all rising trials. All BOLT sitting trials include hand–seat contact.

b) Continuous support transitions: The 31.3 s task starts seated; the robot rises, walks, and ascends stairs while the human walks on level ground. One controller retains continuous state/history without intermediate resets, policy switching, or synthetic prefix/suffix. BOLT completes 9/10, 10/10, and 10/10 trials at increasing heights, versus none for SONIC. All failures begin at stair interaction.

Force reconstruction reproducing one nominal 10 cm episode shows physical load transfer. Phase-mean vertical reaction, normalized by body weight, shifts from 0.628 through the seat to 0.991 through ground-contacting feet after rising, then 0.999 through stair-contacting feet during top-level activity. This single-episode analysis retains the original additional-contact classification.

c) Real-world demonstrations: Figure 1(a) shows the corresponding hardware sequence, ending with an arm gesture. Separate demonstrations show stair ascent and sitting on an office chair and a rigid support (Fig. 1(b–d)). All hardware demonstrations use SMPL motion commands. These provide real-world evidence; Table III reports simulation replays of recorded commands.

d) Climbing extension: One policy continuously executes walking, stair ascent/descent, and table climbing through its SMPL interface in a 32.46 s simulation. Recorded flat walking drives ascent; paired human commands drive descent and climbing, joined offline with a 1 s descent-pose hold. State/history remain continuous. Hand and sustained knee support give way to stable bilateral foot support for the final 1.86 s on the 57.1 cm tabletop (Fig. 4). Geometric body-patch guidance produces this trace; the integrated load-aware refinement (Sec. III-C) is validated qualitatively, without isolating individual reward terms.

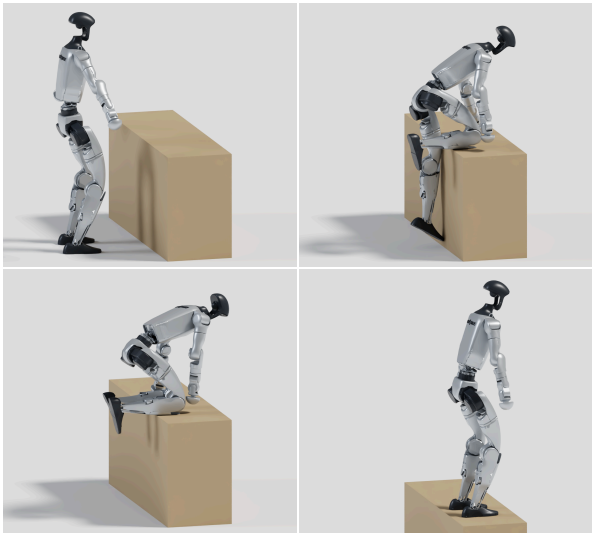


Fig. 4. **SMPL-conditioned table climbing.** Simulation task illustration, read left to right, top to bottom: approach, ascent with hand and knee support, bilateral kneeling, and tabletop standing.

TABLE V

FLAT-GROUND G1 MOTION TRACKING.

Method	No fall ↑	World MPJPE ↓	Root-rel. MPJPE ↓	Root pos. error ↓
SONIC	**10/10**	158.20	**28.26**	153.15
GRAIL	8/10	1768.99	93.86	1786.90
BOLT	**10/10**	**133.43**	34.77	**127.67**

Protocol. Native IsaacLab; the same ten G1 references, initialization, and evaluation code. One full-duration rollout per motion and method; all errors are in mm, averaged equally over motions.
Inputs. GRAIL receives a 121D analytic ground-plane height map and 99D static neutral-object features. No fall denotes full-duration execution with pelvis height never below 0.5 m.

E. Retention of Flat-Ground Tracking

Table V uses four gestures, three walks, and three dances. Mean per-joint position error (MPJPE) covers 14 body positions and all frames, including after falls, with equal motion weighting. Root-relative MPJPE removes root translation without heading alignment.

SONIC and BOLT execute all ten references without falls; GRAIL executes eight in this released-model configuration. BOLT reduces world/root-position error but increases root-relative error versus SONIC. Human-command flat trials have no falls for either model; final-second support/stabilization with root progress ≥ 1.9 m holds in 6/10 BOLT and 10/10 SONIC trials.

F. Feedback Diagnostic: Online Target Updates

We freeze the PD-target sequences from three preselected nominal 10 cm stair rollouts and replay them at 5/10/15 cm, retaining low-level PD feedback. Fixed-target replay reproduces all three nominal trajectories but completes none of the six changed-height trials; closed-loop BOLT completes five of six. Reusing RQ1 controls, the nine replays demonstrate

the role of online target updates, without isolating a latent pathway or training reward.

V. Conclusion

BOLT connects motion intent to physical support through paired data and privileged supervision. SCMP improves construction consistency; force analysis illustrates load transfer, while tracking and feedback diagnostics reveal accuracy tradeoffs and the role of online target updates.

Evaluation covers two held-out human recordings. Alignment sensitivity and extra contacts at higher stairs remain limitations; isolating training objectives requires matched training comparisons.

Blind deployment avoids terrain acquisition and scene-sensor calibration, retaining human capture and state estimation. Future work will connect high-level generators through motion tokens, following MotionWAM [14], and compare end-to-end latency.

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